

Paper 1 — Cross-contamination Audit

*Cancer dark matter scope verification — 38-page audit 의
advisor briefing 버전*

*BRAF/RAS-negative PTC dark matter (~23%) — fusion-driven, epigenetically silenced cancer mechanism 만 다루는 Paper 1. 본 audit 은 **3 source file (Title candidates · Abstract · Outline)** 에 Paper 2/3 territory term 이 누출되었는지 grep 한 결과로, Title Cand 1 ★ + Abstract body 는 PASS 이나 **Outline §2.4b · §2.5 · §3.3 · §3.4(iii)** 에 prose-propagation risk 가 있는 HARD violation 4건 발견. 본 brief 는 advisor 미팅 이전 한 페이지 검토용 single-file scholarly webpage.*

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Glossary

advisor 친화 약어 사전 — 본 *brief* 에서 사용되는 8개 핵심 용어의 정의.

HARD VIOLATION

Source file 의 production content cell (Results · Discussion 행) 에 등장한 forbidden term — drafting 시 prose 로 흘러갈 위험 있음. **advisor 승인 후 mechanical edit 필수.**

SOFT / META VIOLATION

Working-doc 분석 cell (commentary · review-focus note · risk-flag table) 에 등장한 term — 최종 manuscript 에 들어가지 않으나 archive sanitize 권장. 본인 voice 영향 없음.

FORBIDDEN TERM LIST

Paper 2/3 territory 15개 단어 (advisor 2026-05-04 지시): Paper 2 = Hashimoto · HT · thyroiditis · lymphocytic infiltration · autoimmune-overlap · autoimmune-PTC · PTC+HT · DPB1*05:01 ; Paper 3 = Graves · GD · TSAb · hyperthyroid · PTC ≠ GD ; cross-paper = Hashimoto-overlap · Pan-Asian autoimmune-thyroid .

ALLOWED EXCEPTION

Cancer immune-microenvironment 의 일반 관찰은 OK — 예: BRAF V600E HIGHER HLA-I (Bradley 2010 contradict, Discussion 3.1 의 counter-intuitive finding). 단 autoimmune framing 은 제외. **"Immune-overlap subtype"** 은 Paper 2 hook phrase 로 §2.4b take-home 에서만 허용.

A1 – A7 ACTION ITEMS

Audit 결과의 advisor 승인 게이트 7개. A1–A4 HARD (production cell), A5 MED (cohort 결정), A6–A7 SOFT (meta-doc sanitize). § 9 표 참조.

PILLAR 5 BOUNDARY

Paper 1 hook = 5개 pillar 중 마지막. Paper 2 backbone 으로 reserve. Paper 1 §2.4b 는 "older + immune-hot DM1 sub-B is mechanistically distinct" 까지만 진술하고 "warranting separate inquiry" → Paper 2 hook.

TIERA67 · DM1 · DM2

TIERA67 = 67-gene candidate pool (driver gene 미포함, biology-curated); DM1 / DM2 = TIERA67 expression 으로 leiden K=2 unsupervised cluster — DM1 = MAPK/de-differentiation, DM2 = thyroid-lineage-preserved.

8-GENE RAI PANEL

SLC5A5 (NIS) · TPO · TG · TSHR · PAX8 · NKX2-1 · FOXE1 · DIO1 — Yoo 2016 PLOS Genet RAI biology 의 8 thyroid-differentiation core gene.

BRAF/RAS-NEGATIVE DARK MATTER

~23% of PTC — BRAF V600E 음성, RAS hotspot 음성. Xing 2014 NEJM 의 "dark matter" 명명. Paper 1 의 sub-stratification target. 76.8% 가 fusion-positive (RET / NTRK / ALK / BRAF) + epigenetic silencing 동반 (TPO Cohen's d = 2.30).

Audit Overview

§ 1

Hook — 왜 audit 가 필요했는가

Three-paper isolated session policy + prose-propagation risk.

2026-05-04 advisor message: "연구가 짬뽕이 되는 현상" 을 차단해 달라는 지시. 본인이 Q1-Q2 에 진행 중인 cancer dark matter paper (Paper 1), Q3-Q4 의 immune-overlap paper (Paper 2), 그리고 별도 trajectory 인 thyroid hyperfunction paper (Paper 3) — 세 paper 간 evidence layer 가 **저자 머릿속에서는 분리되어 있으나 manuscript 본문에서는 boundary 가 흐려질 risk** 가 존재합니다. 특히 Paper 1 의 outline 에 Paper 2 territory statistic (예: **△ Hashimoto-like 40% vs 67%**) 이 inline 으로 적혀 있으면, drafting 단계에서 prose 로 자연스럽게 흘러 들어가 reviewer 가 "이 paper 의 scope 가 무엇인가" 를 묻는 위험이 발생합니다.

본 audit 은 단순 grep 가 아니라 **prose-propagation risk 평가**: production cell (HARD) vs working-doc commentary (SOFT) 를 분리하고, take-home phrase 가 boundary 역할을 하는지 검증하는 작업입니다. Paper 2 / Paper 3 에 동일 정신의 audit 이 별도 session 으로 진행될 예정이며, 본 file 은 **Paper 1 isolated session** 의 산출물입니다.

PROSE-PROPAGATION RISK — 본 AUDIT 의 핵심 우려

Outline 의 statistic cell 은 Results §X 의 prose 로 그대로 변환되는 경향이 있음. 예: **△ Hashimoto-like 40% vs 67% (OR 0.34, p=0.064 trend)** 가 §2.4b row 에 있으면, draft writer 가 "Sub-B was characterized by ... including a 67% **△ Hashimoto-like** infiltration rate ..." 로 자동 prose 화 함. boundary statement 는 옳더라도 statistic 자체가 cell 에 남아 있는 한 누출 risk 는 0 이 아닙니다.

§ 2

Scope + Method

3 file audited · 15 forbidden terms · 3 severity tiers.

2.1 Files audited (3)

#	FILE	LINES	ROLE
F1	00_title_candidates.md	75	3 title candidates + decision matrix
F2	01_abstract.md	73	Cell Press structured 153-word abstract + numerical claims table + review-focus notes
F3	02_outline.md	328	Section ordering · 5 sub-result rows · Discussion 4 row · risk-flag table

2.2 Severity classification

HARD Production content cell (Results · Discussion 행) — drafting 시 prose 로 propagation 위험.
SOFT Working-doc commentary cell (analysis pros/cons · review-focus note) — final manuscript 미반영, archive sanitize 권장. **META** Planning-doc only (mermaid flowchart annotation · risk-flag table) — sanitize 불필요. **PASS** 0 forbidden term in body.

2.3 Allowed exceptions

Cancer immune-microenvironment 의 일반 관찰 — Paper 1 §3.1 Discussion 의 "*BRAF V600E HIGHER HLA-I (counter-intuitive)*" 은 Bradley 2010 contradict 으로 explicit 허용. "*immune-overlap subtype*" phrase 는 §2.4b take-home 의 Paper 2 hook 으로만 허용. 그 외 모든 autoimmune framing 은 Paper 2 territory.

Chapter 2

Audit Results

§ 3

Executive summary ★

3 file × HARD/SOFT/META 격자 + bottom line.

2 / 3 TITLE CAND · PASS <i>Cand 1 ★ + Cand 2 — 0 forbidden term in title body</i>	PASS ABSTRACT BODY <i>Lines 11–19 · 0 forbidden term · Conclusions (b) clean</i>	4 HARD VIOLATIONS · OUTLINE <i>§2.4b · §2.5 · §3.3 · §3.4(iii)</i>	3 SOFT / META <i>Cand 3 commentary · abstract review-focus · risk-flag table</i>
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Table 1 — File × severity 격자

FILE	PRODUCTION HARD	WORKING- DOC SOFT	PLANNING META	BODY PASS
00_title_candidates.md	0	2 (lines 62, 65)	—	★ Cand 1 lock
01_abstract.md	0	1 (line 66)	—	Body lines 11–19 PASS · n=874 boundary risk
02_outline.md	4 (lines 104, 106, 122, 123)	0	2 (lines 31, 56, 250)	§2.4a + §3.1 + §3.2 PASS · §2.4b · §2.5 · §3.3 · §3.4(iii) HARD

BOTTOM LINE

Title Cand 1 ★ + Abstract body 는 PASS — 즉시 lock 가능. Outline §2.4b · §2.5 · §3.3 · §3.4(iii) 4 cell 에 prose-propagation risk 가 있는 HARD violation. Pillar 5 boundary 의 의도 (Paper 1 = "older + immune-hot DM1 sub-B is mechanistically distinct" 까지) 는 take-home line 112 에서 깔끔하나, §2.4b cell (line 104) 자체에 Paper 2 statistic 5개가 inline 으로 남아 있어 enforce 안 됨. **A1–A4 mechanical edit 30분 후 §2.4b · §2.5 · §3.3 · §3.4 prose drafting 진입 가능.**

§ 4

Title check — 3 candidates

00_title_candidates.md · 75 lines · Cand 1 ★ recommended · 0 forbidden in title body.

1



Mechanism-forward — recommended

LINE 24 · PASS · 0 FORBIDDEN · 99 CHARS

"Fusion-driven, epigenetically silenced dark matter in BRAF/RAS-negative papillary thyroid carcinoma"

Cell Press style 의 mechanism-forward title. Xing 2014 NEJM *"dark matter"* framing 을 cancer-context 으로 직접 인용 — scope-clean. 두 mechanism layer (fusion + epigenetic silencing) 가 99 character 안에 모두 수록.

Recommendation

Lock as v3 title. Decision matrix 에서 Cell Reports Medicine ★★★ + Nat Commun ★★★ 평가.

2

Translational-forward

LINE 40 · PASS · 0 FORBIDDEN · 110 CHARS

"An 8-gene RAI panel enables reflex fusion testing and re-induction therapy in BRAF/RAS-negative thyroid cancer"

RET reflex algorithm + HMA + RAI re-induction rationale. "Re-induction therapy" 가 forward-looking — abstract Conclusions (b) 처럼 신중한 framing 필요.

3

Discovery-forward — title body PASS, commentary SOFT

LINE 56 (TITLE BODY) + 62, 65 (COMMENTARY) · TITLE PASS

COMMENTARY SOFT · 109 CHARS

"An 8-gene panel sub-stratifies BRAF/RAS-negative thyroid cancer into fusion-driven and immune-overlap subtypes"

Title body 는 0 forbidden — "immune-overlap" 은 Paper 2 hook phrase 로 허용. 단 commentary cell 2개에 SOFT violation:

SOFT line 62 (Foregrounds cell)

"DM1/DM2 split + R4-2 fusion+/fusion- heterogeneity + Pillar 5 (△ autoimmune-PTC) layer"

SOFT line 65 (Pros cell)

"Immune-overlap" foreshadows Pillar 5 (△ Hashimoto -DM2 axis + GSE286332)

Recommended sanitization (advisor approval 후)

- L62: Pillar 5 [Paper-2-territory] layer → R4-2 immune-overlap teaser layer (Paper 2 backbone reserved)
- L65: "Immune-overlap" foreshadows Pillar 5 (...) → "Immune-overlap" phrase reserves Pillar 5 main story for Paper 2

§ 5

Abstract check

01_abstract.md · 73 lines · body PASS · n=874 cohort boundary risk.

A

Abstract body — lines 11–19

LINES 11–19 · PASS · 0 FORBIDDEN TERM · 153 WORDS

Background / Methods / Results / Conclusions 4 paragraph 모두 forbidden-term 0 건. Conclusions (b) 검증 결과:

*"DM1 is a fusion-driven, epigenetically silenced subtype, providing a clinical sub-stratification algorithm and motivating evaluation of fusion-targeted therapy and **epigenetic-targeted RAI re-induction**."*

HMA + RAI re-induction *rationale* 만 명시 — autoimmune mechanism 미언급.

"epigenetic-targeted" 표현이 핵심: TPO/DIO1/TSHR promoter hypermethylation 으로부터 derive 된 cancer mechanism rationale 임을 명확히 함.  **PASS**.

B

Cohort boundary risk — n=874 GSE286332 inclusion

LINES 15, 49 · MED · ADVISOR 결정 필요

Methods 의 *"Korean cohorts (n=874)"* 숫자 자체는 clean 하나, line 49 numerical-claims table 에 cohort 분해가 적혀 있음:

"Korean cohorts n=874 | D4-P1: K2(235) + Lee(630) + GSE286332-PTC(9)"

GSE286332 은 Paper 2 의 backbone cohort. PTC arm 9명만 포함하더라도, cohort 명시 자체로 Paper 2 territory 와 boundary 모호. § 10 에서 option (a) vs (b) decision 제시.

C

Review-focus annotation — line 66

LINE 66 · SOFT · META-ONLY · 본문 미반영

"Pillar 5 (△ autoimmune-PTC) 는 abstract 에서 의도적 보류 — Paper 2 의 backbone 으
로 reserve."

본인 review-focus note item 3 — 최종 manuscript 에 들어가지 않으나 archive sanitize
권장.

Recommended sanitization

Pillar 5 ([Paper-2-territory term]) 는 abstract 에서 의도적 보류 → Paper 2
sub-stratification axis (immune-overlap) is intentionally deferred

§ 6

Outline check ★ 4 HARD violations

02_outline.md · 328 lines · §2.4b · §2.5 · §3.3 · §3.4(iii) HARD.

2.4a §2.4a · DM1 epigenetic silencing of differentiation machinery

LINE 103 · PASS · 0 FORBIDDEN TERM · PURE CANCER MECHANISM

Promoter hypermethylation Cohen's d table — TPO 2.30, DIO1 1.24, TSHR 1.20, PAX8
0.97, TG 0.86, FOXE1 0.84, NKX2-1 0.63, SLC5A5 NS · mean 8-gene β 0.385 vs
0.253 · methylation **fusion-independent** · 3-layer model. 스코프 깨끗.  PASS.

2.4b §2.4b · Fusion-negative DM1 = immune-overlap subtype (teaser)

LINE 104 · HARD · 5 PAPER 2 STATISTICS INLINE · BOUNDARY LEAKY

Boundary statement 의 의도는 옳으나 ("Paper 1 stops at 'older + immune-hot DM1 sub-B is mechanistically distinct'") — 동시에 같은 cell 안에 Paper 2 territory 통계 5개가 inline 으로 남아 있어 prose 로 propagate 위험.

Cited fragment from outline cell

"Sub-A vs sub-B: age (37 vs 51), fusion% (84.7 vs 57.9), stage III/IV (15.3% vs 44.4%), CD8/IFN- γ low vs high ($d=-0.5$ to -0.6), Δ Hashimoto-like 40% vs 67% (OR 0.34, $p=0.064$ trend) ; Δ Pillar 5 mediation / Δ HLA n=874 / Δ BCR clonal / Δ sub-B NBNR is Paper 2 backbone — Paper 1 stops at 'older + immune-hot DM1 sub-B is mechanistically distinct'"

Recommended sanitization (advisor approval 후 mechanical edit)

- Drop the Δ Hashimoto-like 40% vs 67% statistic
- Drop the inline list Δ Pillar 5 mediation / HLA n=874 / BCR clonal / sub-B NBNR
- Keep cancer-only sub-A vs sub-B contrasts: age (37 vs 51), fusion% (84.7 vs 57.9), stage III/IV (15.3% vs 44.4%), CD8/IFN- γ low vs high $d = -0.5$ to -0.6
- Replace boundary tail with: "Paper 1 stops at 'older + immune-hot DM1 sub-B is mechanistically distinct'; the immune-overlap mechanism is reserved for Paper 2."

2.5 §2.5 · Cross-cohort + Reflex testing algorithm

LINE 106 · HARD · COHORT + STATISTIC BOUNDARY LEAK

Cited fragment

"... Korean K2/Lee/GSE286332 n=874 + GSE213647 n=632 Δ (28% Hashimoto-like) ; DM1 captures 81.8% TCGA RET+ $\rightarrow$ reflex algorithm; population estimate 48 selpercatinib-eligible per 1000 PTC ..."

Recommended sanitization

- Drop Δ (28% Hashimoto-like) parenthetical
- Resolve GSE286332 inclusion — § 10 option (a) vs (b)

3.3 §3.3 · East-Asian generalizability + Pu 2021

LINE 122 · **HARD** · PAN-ASIAN FRAMING 부적절

Cited fragment

"... 4,300+ East Asian PTC tumors covered; Korean DM 37.8% vs TCGA 28.4%; K2 NBNR mixed phenotype (vascular invasion); △ Pan-Asian Hashimoto-like replication (Korean GSE213647 22-28% ≈ TCGA 18-20%); single-cell external validation in 6 PTC patients (Pu et al 2021 Nat Commun) ..."

△ Pan-Asian Hashimoto-like replication 은 Paper 2 의 generalizability 근거이지 Paper 1 의 cancer dark matter generalizability 근거 아님.

Recommended sanitization

Drop the △ Pan-Asian Hashimoto-like sentence entirely. "Korean DM 37.8% vs TCGA 28.4%" 만으로 cancer-only generalizability 충분.

3.4 §3.4(iii) Methodological limitations

LINE 123 · **HARD** · HLA DANGLING + GSE286332 REFERENCE

Cited fragment

"(iii) Methodological: △ HLA d 1.5+ partial autocorr ($\Delta d=0.30$); K2 mini-index calibration mismatch (R4-4) propagates △ GSE286332 reference."

Paper 1 §2 Results 에 HLA 항목이 없으므로 HLA limitation 은 dangling — anchor 없음.
GSE286332 reference 명시는 Paper 2 territory.

Recommended sanitization

- Drop the HLA d 1.5+ partial autocorr clause entirely (no §2 anchor)
- Replace "propagates [Paper-2-cohort] reference" → "K2 mini-index calibration mismatch (R4-4) — same calibration is propagated through any cohort built on the K2 reference; cross-Korean cohort findings are interpreted with this caveat." (avoids naming the Paper 2 cohort)

3.1 §3.1 · Three-layer DM1 pathology + Landa 2016 dial-back

LINE 120 · PASS · 0 FORBIDDEN TERM · CANCER PAIRED-TRAJECTORY

L1 genetic (76.8% fusion+) · L2 mechanism heterogeneity (sub-A vs sub-B) · L3 epigenetic (TPO d=2.30 fusion-independent). Landa 2016 JCI 126(3):1052-1066 PDTC → ATC continuum framing 으로 DM1 = upstream signature of dedifferentiation trajectory — cancer paired-trajectory, scope-clean.

Bradley 2010 contradict: *BRAF V600E HIGHER HLA-I (counter-intuitive, R3 finding 14)* — explicit **allowed exception** per spec. Cancer immune-microenvironment 의 일반 관 찰이며 autoimmune framing 아님.

3.2 §3.2 · ATA 2015/2025 + Wirth 2020 LIBRETTO-001

LINE 121 · PASS · 0 FORBIDDEN TERM · CANCER THERAPY TRANSLATIONAL

ATA 2015 (Haugen 2016) → ATA 2025 (Ringel 2025) BRAF V600E sole molecular risk modifier 한계 + DM1 reflex algorithm = orthogonal molecular axis; 57% DM1 actionable; LIBRETTO-001 (Wirth 2020 NEJM, ORR 79% in RET-fusion+) — DM1 RNA-score = upstream candidate pool. HMA + RAI re-induction rationale = TPO/DIO1/TSHR promoter hypermethylation 에서 derive (epigenetic mechanism, autoimmune mechanism 무관). "Evaluation, not trial" dial-back tone 유지.  PASS.

Chapter 3

Pillar 5 Boundary Analysis

§ 8

Pillar 5 boundary clarity

5 boundary phrase locations · take-home (line 112) clean · §2.4b cell (line 104) leaky.

Paper 1 vs Paper 2 의 boundary 의도는 source file 5곳에 명시되어 있음. 의도 자체는 일관되나, 의도와 production cell 의 통계 listing 사이에 gap 이 있어 enforce 안 됨.

Table 2 — Pillar 5 boundary phrase locations across source files

FILE:LINE	PHRASE / CONTEXT	DIRECTION	STATUS
02_outline.md:31	"Pillar 5 main story (HLA n=874 mediation, BCR/TLS, sub-B NBNR) 는 Paper 2 reserve. Paper 1 에서는 fusion- DM1 = older + immune-hot 까지만 teaser."	Meta	META
02_outline.md:56	"2.4b Fusion- DM1 = immune-overlap teaser ... Pillar 5 hint · Paper 2 reserve" (mermaid annotation)	Meta	META
02_outline.md:104	"Paper 1 stops at 'older + immune-hot DM1 sub-B is mechanistically distinct'" — boundary tail · BUT 5 Paper 2 statistics inline above	Production	LEAKY
02_outline.md:112	"...with fusion-negative DM1 representing a mechanistically distinct immune-overlap subtype warranting separate inquiry." → Paper 2 hook	Take-home	CLEAN
02_outline.md:250	"Pillar 5 Paper 1 vs Paper 2 boundary 모호 ... 2.4b 마지막 sentence ... → Paper 2 hook 명시. Paper 2 backbone reserve: ..." (risk-flag table)	Meta	META

VERDICT – BOUNDARY 의도 VS ENFORCEMENT GAP

Take-home (line 112) "warranting separate inquiry" 는 깔끔한 boundary phrase. 그러나 같은 §2.4b row 의 "Key claims" cell (line 104) 안에 Paper 2 statistic 5개 (Hashimoto-like 40% vs 67% , Pillar 5 mediation , HLA n=874 , BCR clonal , sub-B NBNR) 가 inline 으로 listed — drafting 단계에서 prose 로 propagate 가 자동 발생. Line 104 sanitize (A1) 시에만 boundary 가 enforced 됨.

A1 – A7 advisor approval gate

7 mechanical edits · A1–A4 HARD blocking · A5 MED · A6–A7 SOFT · 총 30분 post-approval.

A1

02_outline.md:104 · §2.4b key-claims cell

Drop 4 Paper 2 inline statistics (△ Hashimoto-like %, △ Pillar 5 mediation, △ HLA n=874, △ BCR clonal, △ NBNR mention list); keep age / fusion% / stage / CD8-IFN-γ contrasts + "warranting separate inquiry" boundary tail.

HARD

~5 min

A2

02_outline.md:106 · §2.5 cross-cohort cell

Drop △ (28% Hashimoto-like) parenthetical; resolve GSE286332 inclusion question (§ 10 option a vs b).

HARD

~3 min

A3

02_outline.md:122 · §3.3 East-Asian generalizability

Drop △ "Pan-Asian Hashimoto-like replication" sentence; "Korean DM 37.8% vs TCGA 28.4%" 만으로 cancer-only generalizability 충분.

HARD

~2 min

A4

02_outline.md:123 · §3.4(iii) Limitations

Drop HLA d 1.5+ partial autocorr clause (no §2 anchor in cancer-only Paper 1); rephrase GSE286332 reference 를 generic K2-calibration caveat 로 — 외부 cohort 명시 회피.

HARD

~5 min

A5

01_abstract.md:15, 49 · n=874 cohort decomposition

Cohere with A2 결정. Option (a) 유지 + STAR Methods caveat 또는 Option (b) n=865 (K2+Lee only)
— § 10 참조.

MED

~3 min

A6

01_abstract.md:66 · review-focus note item 3

Replace Pillar 5 ([forbidden-term]) → Paper 2 sub-stratification axis (immune-overlap). Archive sanitize · 본인 voice 영향 0.

SOFT

~2 min

A7

00_title_candidates.md:62, 65 · Cand 3 commentary cells

Sanitize Foregrounds + Pros cells of Discovery-forward candidate. Title body 자체는 PASS · sanitize
는 archive cleanup 만.

SOFT

~2 min

TOTAL MECHANICAL-EDIT TIME POST-APPROVAL

~22 분 (A1 5 + A2 3 + A3 2 + A4 5 + A5 3 + A6 2 + A7 2). Advisor 가 A1–A4 (HARD) + A5
(option a/b) 결정만 confirm 하면 본인 voice 영향 없는 mechanical edit. 이후 §2.4b · §2.5 ·
§3.3 · §3.4 prose drafting 가능.

§ 10

GSE286332 PTC(9) decision — option (a) vs (b)

A2 + A5 의 cohort boundary resolution · 본 audit 의 유일한 판단 prompt.

Abstract line 49 + outline line 106 의 Korean cohorts $n=874 = K2(235) + Lee(630) + GSE286332-PTC(9)$ 분해에서 GSE286332 PTC arm 9명을 어떻게 처리할지 advisor 결정 필요.

OPTION A

Keep $n=874$ + STAR Methods caveat

NUMBERS	K2 235 + Lee 630 + GSE286332-PTC 9 = 874
METHODS	STAR Methods 1 줄 추가: <i>"Only the PTC arm of GSE286332 is included; the [Paper-2-cohort-arm] is reserved for Paper 2."</i>
PROS	Statistical power 보존 · $n=874$ round 숫자 유지 · 모든 figure caption 그대로
CONS	Paper 2 cohort 명시로 boundary 모호 risk · reviewer 가 "이 cohort 가 Paper 2 에도 사용?" 우려 가능

OPTION B ★

$n=865$ (K2 + Lee only)

NUMBERS	K2 235 + Lee 630 = 865
METHODS	GSE286332 cohort 전체 Paper 2 territory 로 reserve · Paper 1 STAR Methods 에서 미언급
PROS	Cancer-only scope 절대 깨끗 · Paper 2 backbone 무관계 · reviewer Q 사전 차단
CONS	1.0% statistical power loss (9/874) · n round 숫자 변경 (모든 figure caption 1 line edit)

★ **RECOMMENDED** – cancer-only scope 강화

DECISION NEEDED – ADVISOR 한 줄 답

Option (a) "*keep $n=874$* " 또는 Option (b) "*recompute $n=865$* ". **본인 권장: option (b)** — 1% power loss 는 stat 영향 미미하나 boundary 명확화의 가치가 큼. 단 figure caption 의 n 표기 일관 변경 필요 (~10 분 추가).

Next Steps

§ 11

Post-approval workflow

5/4 advisor 답장 → 5/5 prep day → 5/6 Hook + Title lock → 5/7–6/13 marathon W1–W6.

2026-05-04 AM Audit 완료 — *cross_contamination_check_paper1.md* (214 lines) + 본 webpage 산출

3 source file × 15 forbidden term · 4 HARD + 3 SOFT/META violations 식별 · Pillar 5 boundary 4 location verified · A1–A7 action items 정리.

2026-05-04 PM Advisor 답장 + A1–A4 confirm · A2/A5 option (a) vs (b) 결정

본 brief + *cross_contamination_check_paper1.md* 전송 · advisor 한 줄 답 (e.g. "A1–A4 confirm / A2 option B / A6–A7 sanitize 진행") 으로 mechanical edit 권한 획득.

2026-05-04 PM (cont.) Outline + Abstract + Title-candidates mechanical edit (~22분)

A1–A7 적용 · Claude Code 가 본인 voice 보호 영역 (§2.4b prose · Hook 첫 단락 · Limitations) 미접근 · 단순 cell content sanitize.

2026-05-05 본인 prep day — Landa 1.5h + Yoo 30min + ATA 1–1.5h

Discussion 3.1 (Landa 2016 paired continuum) 인용 verify · Yoo 2016 PLOS Genet methods + reverse-causality 차단 review · ATA 2015 + 2025 fusion/epigenetic 반영 여부 시나리오 1/2/3 분기.

2026-05-06 Title Cand 1 ★ lock + Hook 첫 단락 본인 키보드

"Fusion-driven, epigenetically silenced dark matter in BRAF/RAS-negative papillary thyroid carcinoma" 확정 · Introduction §1.1 hook (sprint vs marathon mode 의 voice-protected 영역) 본인 작성.

W1–W2 Methods + Suppl freeze → W3–W4 Results + Discussion draft → W5–W6 cover letter + submission. 본인 voice 보호 영역 (§3.1 Discussion 첫 paragraph · §3.4 Limitations) 본인 키보드 · 나머지 scaffolding · figure · Methods 정리는 Claude Code 보조.

VALIDATION CHECKLIST – 본 WEBPAGE 자체

- ☒ forbidden term grep 0건 (allowed quote wrapper `<span class="forbidden-quote">` 외 plain text 등장 0)
- ☒ Paper 2 brief webpage design language 일관 (Cormorant Garamond + Newsreader · Burgundy/Gold/Sage palette · stat-grid · callout · viol-card · timeline)
- ☒ Action items table A1–A7 severity badge 정확 (HARD 4 · MED 1 · SOFT 2)
- ☒ Pillar 5 boundary 5 location 정확 인용 (lines 31, 56, 104, 112, 250)
- ☒ Print-friendly @media print rules · break-inside:avoid 적용
- ☒ 한국어 main + 학술 영어 mixing — 자연스러운 phrase boundary
- ☒ Single file · 외부 dependency 0 (CSS inline · Google Fonts CDN 1개만)

Colophon

Set in Cormorant Garamond (display) + Newsreader (body) + JetBrains Mono (technical).
Palette: Burgundy 7B1F2A · Gold B8893C · Sage 5C7B6E · Ink 0F1A2E · Paper F4EFE6.
Single HTML file · CSS inline · 0 JS · print-optimised.

Source & provenance

Source · `project/manuscript_v8/cross_contamination_check_paper1.md` (214 lines).
Source files audited · `00_title_candidates.md` , `01_abstract.md` , `02_outline.md` .
Audit policy · advisor 2026-05-04 isolated-session directive.

Scope & boundaries

Paper 1 · cancer dark matter (BRAF/RAS-negative PTC).
Paper 2 / Paper 3 territory terms · 0 plain-text occurrences (15 quote-wrapped).
본인 voice 보호 영역 (§3.1 첫 paragraph · §3.4 Limitations · Hook 첫 단락) 미접근.